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The effects of energy use on infant mortality rates in Africa

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ABSTRACT

The availability and supply of electricity remains pivotal to the social and economic wellbeing of people and society. Researchers have confirmed that access to electricity greatly impacts quality of life as well as life expectancy. Today, while most developing nations continue to grapple with the consequences of an epileptic power supply and its effects on their socio-economic and environmental prosperity, developed nations are focused on promoting policies and programs to sustain and improve the provision of adequate electricity that guarantees less emissions, pollution and hazards to human health. At the moment, about a hundred million people lack access to enough clean, affordable and reliable energy to meet their daily needs. However, the growing need for electricity to power our world and our lives portends dangerous consequences that must be prevented. This paper examines the effects of energy use on the infant mortality rate in Africa by using a panel of 23 African countries for the period from 1999 and 2014. The Gary Becker hypothesis and the Grossman models are used to examine the relationship between energy related predictors and infant mortality fundamentals. The results of the study show that energy predictors have a negative and significant impact on infant mortality rates among the African countries examined. Further results show that a high degree of pollution causes a rise in mortality rates, whereas proceeds from natural resource rents reduce the level of infant mortality. Thus, this empirical estimation suggests that there is an urgent need for government intervention programs to rescue the region from this energy dilemma.

1. Introduction

The demand for energy continues unchecked, but it has also been reshaped by environmental concerns. Indeed, a huge amount of resources has been channeled into finding carbon mitigating technologies (Tan et al., 2018). Despite the growth in the literature on energy use in Africa, it appears that the existing research is still in its infancy. One important question that remains is whether the infant mortality rate depends on the quality and quantity of energy use, or perhaps whether energy shortage is an important predictor of one's life expectancy. This paper examines the impact of energy use on infant mortality rates in Africa. The study contributes to the growing search for environmental regulations and policies needed to enhance energy efficiency and protect public health (Figs. 1 and 2).

At some point, possibly in 2009, the persistent increases in healthcare expenditure became a matter of discourse in the academic realm. Certainly, the price of food related items was also inflated simultaneously, but the major goods attached to life was seriously interrogated (Alola et al., 2019; Nadimi and Tokimatsu, 2018b; Shobande and Shodipe, 2019). It would appear that when energy related crises are connected with health-related matters, the subject of enquiry becomes existential and the literature on energy misery especially valuable. This is because the quantity and quality of energy consumed is to some extent predictive of the likelihood of an individual's life expectancy (Nkalu and Edeme, 2019; Satre-Meloy et al., 2019).The present energy crisis is driven by coal and gas, which has continued to dominate the fuel mix. These sources have proven to be flexible and highly competitive economically compared to alternative sources of energy. The challenge is that the burning of coal and natural gas involves emitting CO_2 which is a driver of greenhouse effects and, consequently, a threat to the survival of the human race and unborn generations (Best and Burke, 2018; Nathaniel and Iheonu, 2019).

Obviously, one of the major dividends of healthy people is a more productive workforce, as healthy infants and children can develop better and become productive (Colman and Dave, 2018; Dávila-Cervantes and Pardo-Montaño, 2018; Grisolía et al., 2018; Markowitz and Kaestne, 2018). Of course, realizing such dividends cannot be achieved given the present state of health of the people in the region. In fact, recent reports have confirmed that the essential health facilities needed to preserve and extend human life expectancy are lacking (Rich and Collins, 2018; Satre-Meloy et al., 2019). There are no doubt burgeoning challenges for the region. First, there is an inadequate supply of electricity in the region.

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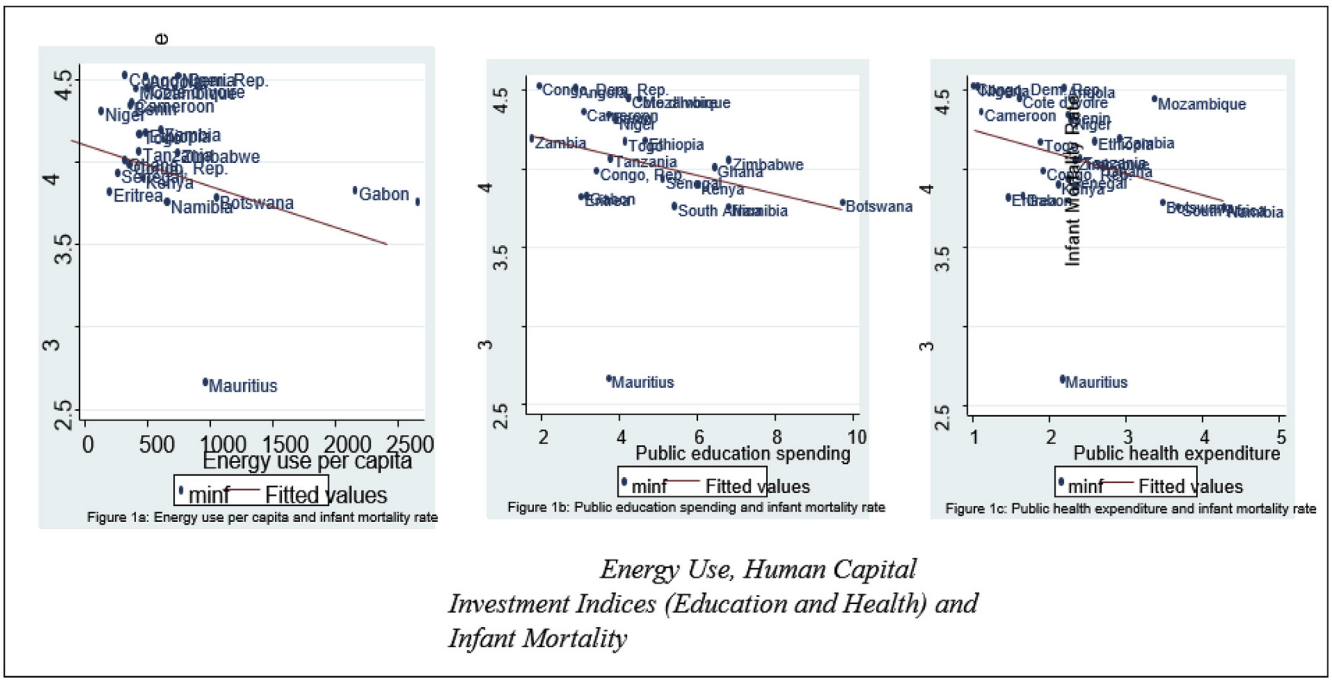

Fig. 1. (a–c) energy use, human capital investment indices (eductation and health) and infant mortality.

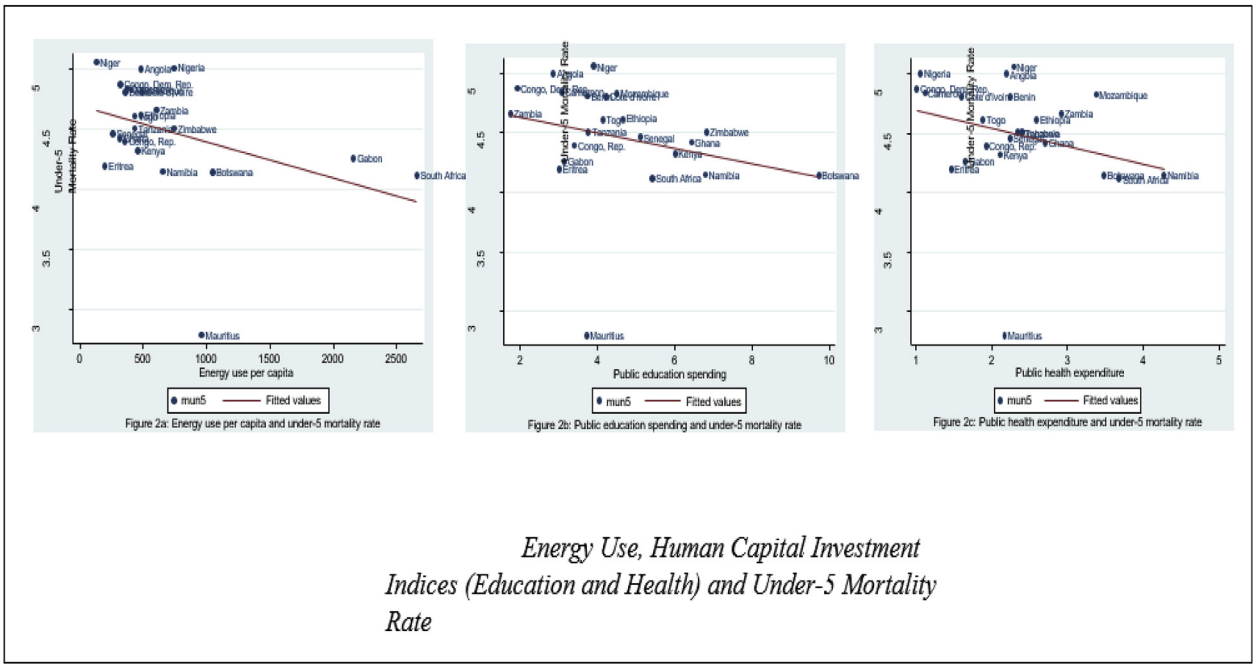

Fig. 2. (a–c) energy use, human capital investment indices (eductation and health) and under −5 mortality rate.

This has resulted in extensive public electricity supply that is highly polluting (Iwayemi, 2008; Nepal and Paija, 2019). Second, the outcome of this shortage of electricity results from; a decay in existing infrastructure, a lack of international commitments, a poor regulatory framework, a lack of regional electricity coordination, and difficulties in accessing financing for projects. These factors contribute in no small measure to an impact on the quality of life and the infant mortality rates (Jawadi et al., 2018). This lack of access to clean electricity is one of the factors that has led to an increase in the mortality rate. Available evidence has shown that the leading cause of a higher mortality rate is exposure to non-communicable diseases, which are possibly associated with the quantity and quality of energy consumed (Aristondo and Onaindia, 2018; Jawadi et al., 2018).

The sufficient supply of consumable goods, both quantity and quality, at affordable prices is an essential prerequisites for preserving and extending life in the region (He and Li, 2019). The current increase in population also shows that the demand for healthcare has become a profitable venture in Africa, since willingness to pay for food consumption and health care access are relatively high and beyond most individuals' wages (Sadath and Acharya, 2017; Shobande, 2019).

Table 1
Descriptive statistics.

Variables	Measurements	Mean	Std Dev.	Maximum	Minimum	Obs.
MINF	Mortality rate, infant (per 1000 live births)	63.520	23.833	145.6	13.1	368
	Mortality rate, infant (per 1000 live births) (log)	4.070	0.435	4.981	2.573	368
MUN5	Mortality rate, under-5 (per 1000 live births)	99.837	42.889	265.6	14.7	368
	Mortality rate, under-5 (per 1000 live births) (log)	4.494	0.515	5.582	2.688	368
EC	Energy use (kg of oil equivalent per capita)	661.98	608.52	3098.4	113.42	361
GEDU	Government expenditure on education, total (% of GDP)	4.418	1.628	10.679	1.100	207
HEXP	Health expenditure, public (% of GDP)	2.293	1.031	5.357	0.045	366
GDPPC	GDP per capita (constant 2010 US$)	2312.6	2630.5	11925.8	186.66	365
	GDP per capita (constant 2010 US$) (log)	7.170	1.067	9.386	5.229	365
CO2	CO_2 emissions (metric tons per capita)	1.076	1.921	10.024	0.017	365
TNAT	Total natural resources rents (% of GDP)	14.344	14.172	62.693	0.001	365
INSQ	Institutional Quality	−0.552	0.662	0.880	−2.100	368
FEDU	School enrollment, secondary, female (% gross)	40.498	27.165	111.52	4.488	209
ISF	Improved sanitation facilities (% of population with access)	31.064	20.071	93.2	3	367
IWS	Improved water source (% of population with access)	67.334	17.239	99.9	21.4	367
INF	Inflation, consumer prices (annual %)	101.86	1335.50	24411	−8.484	344
URBAN	Urban population growth (annual %)	3.676	1.240	5.744	−0.351	365

Note: Std. Dev. is standard deviation; Obs. is observation; and GDP means gross domestic product.
Source: Investigator Computation, 2019.

Table 2
Correlation matrix.

	MUN5	EC	GEDU	HEXP	GDPPC	CO2	TNAT	INSQ	ISF	IWS	INF	FEDU	URBAN
MINF	0.985	−0.358	−0.260	−0.297	−0.555	−0.407	0.293	−0.586	−0.616	−0.662	0.030	−0.732	0.580
MUN5	1	−0.364	−0.256	−0.286	−0.558	−0.411	0.275	−0.559	−0.632	−0.667	0.031	−0.746	0.592
EC		1	0.113	0.245	0.680	0.884	0.029	0.371	0.533	0.547	−0.002	0.766	−0.373
GEDU			1	0.493	0.151	0.169	−0.238	0.283	0.059	0.285	−0.078	0.332	−0.053
HEXP				1	0.240	0.312	−0.294	0.566	0.164	0.262	−0.102	0.387	−0.146
GDPPC					1	0.659	0.065	0.539	0.700	0.794	−0.032	0.854	−0.410
CO2						1	−0.094	0.482	0.584	0.553	−0.015	0.755	−0.435
TNAT							1	−0.504	−0.185	−0.271	0.032	−0.333	0.347
INSQ								1	0.510	0.614	−0.111	0.690	−0.450
ISF									1	0.625	0.014	0.772	−0.675
IWS										1	0.018	0.858	−0.656
INF											1	−0.080	−0.110
FEDU												1	−0.692

Note: MINF denotes infant mortality(per 1000 live births); MUN5 represents under-5 mortality rate (per 1000 live births); EC is energy consumption; GEDU is government expenditure on education; total (% of GDP); HEXP denotes health expenditure, public (% of GDP); CO2 is carbon emission; TNAT is total natural resources rents (% of GDP); GDPPC is GDP per capita (constant 2010 US$); INSQ is institutional Quality; INF means inflation, consumer prices (annual %); ISF is improved sanitation facilities (% of population with access); IWS denotes improved water source (% of population with access); FEDU denotes School enrollment, secondary, female (% gross); and URBAN is urban population growth (annual %).
Source: Investigator Computation, 2019.

However, several billions of dollars of public investment into capacity expansion in the energy industry conflicts sharply with the extremely poor supply outcome measured by refinery output, rise in imported fuel, and frequent power outage and voltage disparities (Shobande and Shodipe, 2019).

The rising price of food in Africa has varying levels of influence on an individual's health. First, the increase in the price of food will affect domestic prices depending on the degree of volatility in the international food prices, price regulation, and structure of the market for consumer goods (Dyjack, 2018). Equally, studies have shown that the effects of rising food price affects the quality of life through inflationary pressure, which results in an inability to pay and, consequently, chronic poverty (Bouzarovski, 2017; Robinson et al., 2018). Beyond the obvious, the instant feedback mechanism of the rising effects can lead to mental health issues, internal frustration and socio-demographic pains (Holman et al., 2018).This is especially the case when there is a 3-digit inflationary pressure that might possibly lead to a decline in health status (Kiviniemi et al., 2018).

For nearly a decade, the electricity sector has been thought of as a natural monopolistic industry where government is seen as the sole supplier of electricity for residential households and industry. The economical and efficient production and distribution of electricity do not only depend on the reliability of private and public partnerships but are also dictated by a monopoly of the government, who regulates the price mechanism of entry, the nature of service quality, and the ways in which firms must behave. This poses a challenge, as there is no Regional Electricity Regulatory Commission which might be expected to provide the legal basis for the operation and the institutional control over competition among the firms in the generation, transmission and distribution of energy and enhance the efficient performance of the sector. It appears that this regulated environment has been compromised, leading to the exploitation of the consumers.

The issue is that the conversion of energy, as well as its benefits, in Africa has been miserable, possibly due to the high capacity to produce, which has not been tapped, or probably lack of commitment by the government, private investors and international donors. These groups appear to lack commitment to the region, likely because the cost of investment is greater than the predicted profit. Substantial impacts have resulted in the mounting death rate of an adult population that would have contributed their quota to the production level of the economy (Yang and Chen, 2014). Moreover, the increasing mortality rate of both born and unborn children also results from the insensitive nature of government towards the energy needs of the infant population. Worse still, the nature of the spread of diseases originating from the shortage of power in Africa and the need to provide solutions are major reasons for the importance of research on these issues.

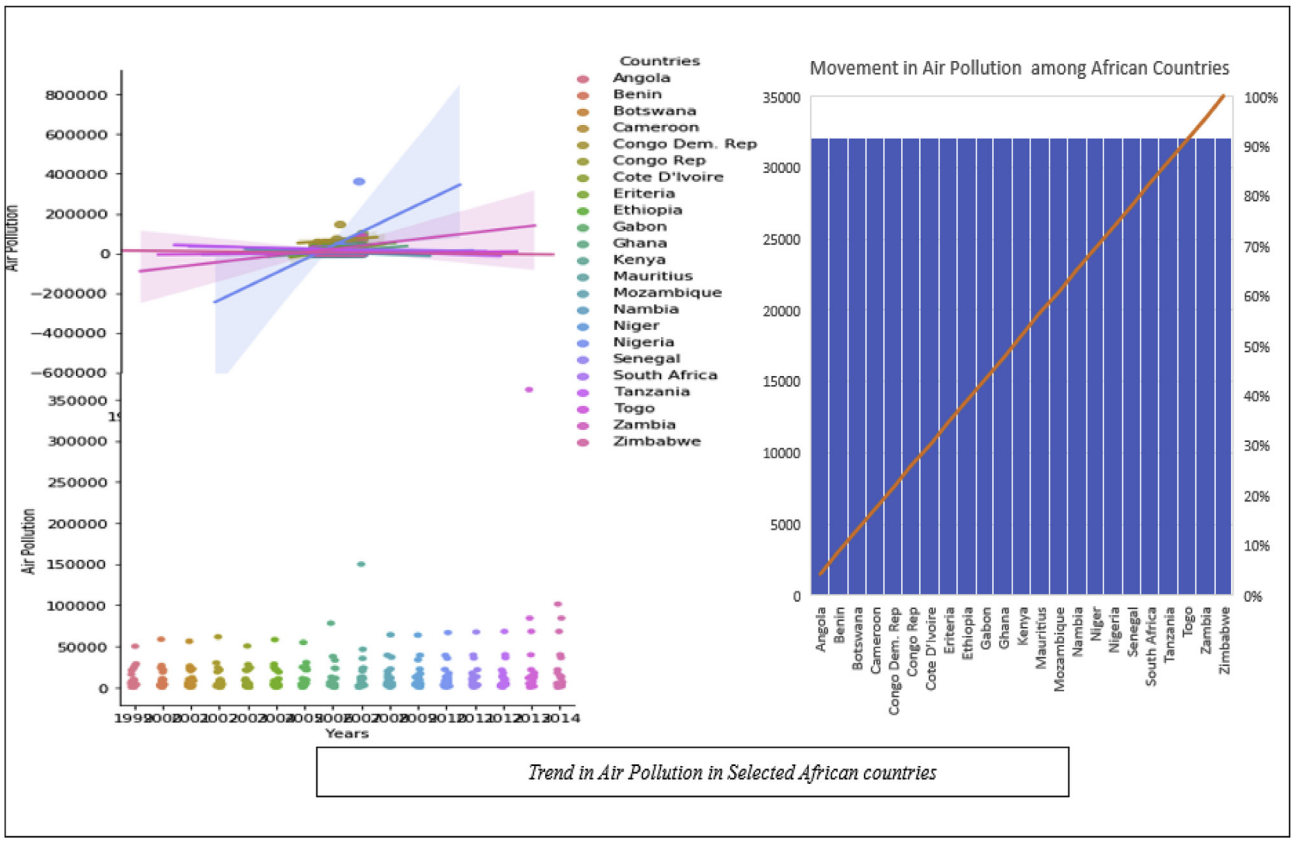

Fig. 3. Trend in Air Pollution in Selected African countries.

The most pressing issue confronting the African continent is generating electricity in a way that is within the capacity of existing facilities (Iwayemi, 2008; Nadimi et al., 2017). The lack of competition and full deregulation, combined with little to no commitment by governments, has led to the increase in the use of environmentally unfriendly energy (Biomass) and the proliferation of an independent energy supply through the use of generators, water purification and captive grid systems, and personalised sewage treatment systems. The poor regulatory mechanism in the region has also led to an unresolved vandalization of pipelines and the destruction of existing power infrastructure (Al-Mulali, 2016; Horbach and Rammer, 2018; Zeng et al., 2018).

Worst yet, these unreliable electricity services, which have resulted from huge technical and non-technical losses, provide support for those who object to paying for unused energy. This, coupled with the harshness of the economic condition, has made it difficult for the survival of common people in the region. Power companies are not helping the situation with their excessive billing, which is a response to usage by unmetered customers and also a result of poor customers' rights regulation (El et al., 2010). They often still employ old technology, such as distribution wire and outdated transformers, leading to poor measurement. All of these issues have had a negative effect on the health of the people. Therefore, the need to examine and reconsider the economic and social cost of the consequences of energy use is not only tied to the peculiarity of the growing population and the disastrous growth that has been recorded so far in the region, but also prompted by the need to quickly address the rising death rate of children under five and the decreasing rate of life expectancy in the region.

The study probes the extent to which the existing health investment could possibly improve life expectancy without compromising the quality of life of, or imposing sacrifices on, unborn generations. The contribution of this study to the extant literature lies in the empirical strategy and measures of variables used, combined with the panel econometric adopted, as well as the peculiarity of the environment of study. First, the dataset covers about 23 African countries and examines the effect of energy predictors on child mortality rate indicators in Africa. The study employs a *system* of generalized method of moment (GMM) econometric techniques. This is the most appropriate methodology, since it has the possibility of predicting the dynamic human condition.

The current study on the relevance of energy uses and infant mortality rate in Africa is motivated by at least five important strands in the recent literature. First, the present and future growth in energy uses is now an inevitable crisis that must be urgently addressed, given the global decrease in the quality of life resulting in a declining rate of life expectancy and an increase in the premature death of younger generations. Second, there is a growing call for scholars and policy makers to consider the relative impact of energy decisions on poor societies that are more likely to suffer the negative effects of increasing or decreasing the supply and access to energy. Third, mixed results have been documented on the correlates of energy uses and quality of life; here, the aim is to reconcile the theoretical understanding of the subject with the empirical facts. This paper contributes to the existing literature on the quality of life in a specific context, as well as with energy and health economics in general. It will be beneficial to a few audiences ranging from the UN, the World Health Organisation (WHO), the Petroleum Technology Development Fund (PTDF) and the Intergovernmental Panel on Climate Change (IPCC), all of whom have been dedicated to sustainable development in sub-Sahara Africa. It will also serve as a point of reference to academics who are currently investigating health and energy related problems.

The study is organized as follows: Section 2 provides a synthesis of the existing literature; section 3 describes the methodology used as well as relevant sources of data; section 4 presents the empirical results and discussion of the findings; section 5 concludes the paper with a workable recommendation.

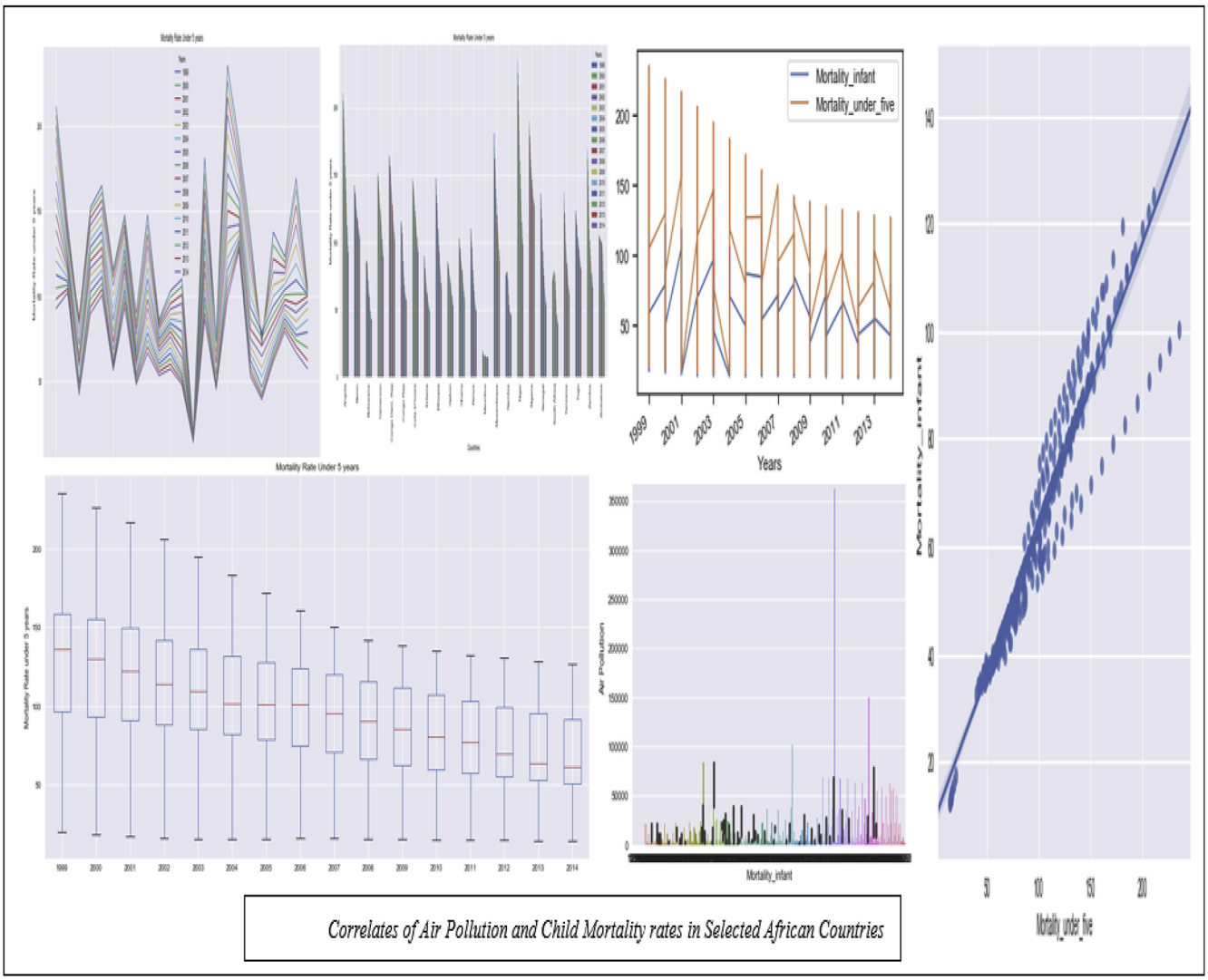

Fig. 4. Correlates of Air Pollution and Child Mortality rates in Selected African Countries.

2. Synthesizing the literature

As far back 1960s, the fields of economics had taken different theoretical directions. The application of economic theories is widely considered in different academic fields ranging from energy, health, environmental related issues. This wide spectrum of theoretical critiques from both environmental and energy related issues have paved the way to evaluate the linked between health and energy sector.

In the extant literature, the standard theoretical framework takes up the possibility of explaining the link between the production and consumption of energy resources as it affects infant health related risks; it is rooted in the Gary production theory. Gary emphasised the difference between health as an output that enters the utility function and health as one of a number of inputs into its production in his review (Galama and Kapteyn, 2009; Hartwig and Sturm, 2018; Li et al., 2019). The empirical modelling strategy in economics that provides the modelling framework for assessing households as both consumers (and producers) is founded on the demand for a healthcare model presented in the seminar work of Grossman and (Becker and Murphy, 1988). The Grossman model highlights health as the most precious good, which suggests that the demand for healthcare is a derived demand; in the model, individuals are both consumers and producers of health. The model predicts that an individual would invest in health until the marginal benefit of health equals its marginal cost; this equilibrium demand for health entails that the length of an individual's life would be determined endogenously.

Empirical testing of the Grossman model has now been integrated into cost analysis to support collective decision making. The intuitive appeal of the so-called welfare economist is that an individual's decision to acquire more and less health is based on personal behavior but constrained by their political context (Shobande and Etukomeni, 2018). This implies that the welfare cost that an individual has enters the collective decision rules exclusively through the quality gained by their society for preserving its health status. While it is commonly assumed that the state of any nation reflects only its health policy, the energy access as well as policy framework for preserving and extending the life of its citizens against resource also figures in deliberation.

Evidence of the effects of energy use on infant health is also inconclusive and problematic. In empirical assessment, two set of theoretical papers are widely contested on whether or not energy uses has a consequential effect on mortality rates. The first strand contends that if there is no link between energy consumption and life, conservative energy policies cannot stimulate or drive quality of life. However, the fact that the conservative hypothesis contends that a link between energy access and quality of life implies that conservative energy policies are required to foster life (Nadimi and Tokimatsu, 2018b).

It is intuitive to believe that higher investment in quality and quantity of electricity energy production has the positive effect of preserving and extending life, given the contest of consumption and production related

Table 3
Baseline pooled regression results (energy consumption and child mortality rate)[a].

Variables	Infant Mortality Rate (MINF)				Under-5 Mortality Rate (MUN5)			
	1	2	3	4	5	6	7	8
Energy consumption	0.016	**−0.114****	−0.015	**−0.139*****	−0.065	**−0.262*****	**−0.100****	**−0.290*****
	(0.047)	**(0.056)**	(0.041)	**(0.049)**	(0.056)	**(0.061)**	**(0.050)**	**(0.054)**
Carbon emission		**0.063*****		**0.061*****		**0.095*****		**0.092*****
		(0.014)		**(0.014)**		**(0.017)**		**(0.017)**
Total natural resource rent			**−0.009*****	**−0.009*****			**−0.010*****	**−0.010*****
			(0.002)	**(0.002)**			**(0.003)**	**(0.002)**
GDP per capita	0.077	0.054	**0.213*****	**0.186*****	**0.131****	0.096	**0.285*****	**0.243*****
	(0.056)	(0.061)	**(0.052)**	**(0.053)**	**(0.066)**	(0.073)	**(0.059)**	**(0.059)**
Govt. Expenditure on education	**−0.042*****	**−0.041****	**−0.047*****	**−0.047****	**−0.049*****	**−0.049****	**−0.056*****	**−0.055****
	(0.016)	**(0.018)**	**(0.017)**	**(0.019)**	**(0.019)**	**(0.022)**	**(0.020)**	**(0.023)**
Health expenditure	0.007	−0.008	0.015	0.000	0.020	−0.002	0.029	0.007
	(0.030)	(0.031)	(0.030)	(0.031)	(0.035)	(0.036)	(0.035)	(0.036)
Institutional quality	**−0.198*****	**−0.246*****	**−0.292*****	**−0.334*****	**−0.184*****	**−0.256*****	**−0.290*****	**−0.354*****
	(0.052)	**(0.053)**	**(0.052)**	**(0.053)**	**(0.061)**	**(0.062)**	**(0.061)**	**(0.062)**
Inflation rate	**0.001*****	**0.001*****	**0.002*****	**0.002*****	**0.001***	**0.001****	**0.001****	**0.0015*****
	(0.0004)	**(0.0004)**	**(0.0004)**	**(0.0004)**	**(0.0005)**	**(0.0004)**	**(0.0005)**	**(0.0004)**
Urban population rate	−0.007	−0.012	−0.015	−0.019	0.006	−0.001	−0.003	−0.010
	(0.038)	(0.039)	(0.034)	(0.035)	(0.044)	(0.047)	(0.040)	(0.042)
Improved sanitation facilities	**−0.013*****	**−0.013*****	**−0.015*****	**−0.015*****	**−0.015*****	**−0.015*****	**−0.017*****	**−0.017*****
	(0.002)	**(0.002)**	**(0.002)**	**(0.002)**	**(0.002)**	**(0.002)**	**(0.002)**	**(0.002)**
Improved water sources	**−0.006****	**−0.005***	**−0.010*****	**−0.009*****	**−0.009*****	**−0.009*****	**−0.014*****	**−0.013*****
	(0.003)	**(0.003)**	**(0.002)**	**(0.003)**	**(0.003)**	**(0.003)**	**(0.003)**	**(0.003)**
Constant	**4.280*****	**5.173*****	**3.965*****	**4.840*****	**5.079*****	**6.428*****	**4.725*****	**6.058*****
	(0.289)	**(0.326)**	**(0.267)**	**(0.320)**	**(0.332)**	**(0.347)**	**(0.321)**	**(0.362)**
Adj. R-squared	0.674	0.701	0.698	0.723	0.682	0.725	0.703	0.744
F-test	**40.38*****	**47.45*****	**50.03*****	**67.82*****	**37.41*****	**55.73*****	**37.54*****	**69.48*****
Observations	197	197	197	197	197	197	197	197

Note: Standard errors in parentheses, while ***, ** and * denote $p < 0.01$, $p < 0.05$, and $p < 0.10$ respectively. The bolded values signify significance of estimated parameters and F-statistics.

[a] Female education was removed because of its high correlation coefficient with mortality rate and the covariates. This led to the omission of the variables itself and other regressors like energy consumption, government spending on education, GDP per capita and institutions under the system GMM approach. The results are available on request.

Source: Investigator Computation, 2019.

factors (Niu et al., 2013). This might seem possible, since an individual produce and consume energy concurrently. Studies have further attempted to explain this scenario in the context of human capital preservation, safe future productivity, and improve wellbeing. A few studies have supported the theoretical proposition of the Becker hypothesis which explains the relationship between consumption related issues and production as it affects human capital investment (Acharya and Sadath, 2019; Clayborne and Colman, 2019). For example, a study by Wang and Luo (2018) examined the spatial relations between life expectancy and different heating energy utilisations in China using a spatial autocorrelation model. It reported that the utilization of biomass has an adverse effect on life expectancy in China. Similarly, Zeleňáková et al. (2018) examined and evaluated the environmental impacts of small hydro power plants by quantitative evaluation, using a matrix and the environmental impacts on the health and social life of the population in the study area. They reported that the negative impact on the environment is minimized in the preparatory phase of the project.

Some studies on energy uses and the quality of life traced the poor health performance and increase in cases of mortality experienced in developed countries to a high level of carbon-intensive energy related pollution (Al-Mulali, 2016). Martins (2005) examined the impact of energy sources on the quality of life in South Africa using a structured questionnaire conducted in the rural area of KwaZulu Natal, one of the nine largest cities in South Africa. The researcher used a sample survey of 404 selected households out of a population of 5900 households. The results show that energy source was a key determinant of the quality of life in the country during the reviewed period.

Mazur (2011) analyses the impact of electricity consumption on the quality of life among 21 industrialized countries between 1980 and 2006 using correlation analysis, and the results show that increases in energy improved the quality of life among countries considered for the analysis during the period reviewed.

Chen et al. (2017) presented a microeconomic-based quantitative analysis scheme to evaluate the energy efficiency of cities based on the quality of life and energy consumption. Representing the quality of life by a utility function, they developed a Constant Elasticity of Substitution model to estimate the individual demand for non-mobility goods, car trips, and public transport at a maximum utility level model for 88 traffic analysis zones included in Nagasaki metropolitan region (Nagasaki, Isahara, Omura) in Japan. Their simulation result indicated that energy consumption is closely related to quality of life.

Al-Mulali (2016) investigated the bi-directional longitudinal relationship between energy consumption and quality of life in 198 countries between 1990 and 2009. They employed canonical co-integrating regression (CCR); the results revealed that energy consumption improves the quality of life in 70% of the countries despite their different income levels.

Nadimi et al. (2017) focused their analysis on the consequence of energy consumption on the quality of life indicators, with a bias for poverty in developed and developing countries between 2003 and 2013. They used the 10 based logarithm data sourced from World Development Indicator. Their study employed the multivariate statistical techniques, factor analysis, and K-mean clustering approach. The results of their study confirmed that different channels of energy consumption impact both positively and negatively on the quality of life indicators.

Nadimi and Tokimatsu (2018a,b) attempted to model the quality of life in terms of energy and electricity consumption for 112 countries between 2005 and 2013, using a panel correlation analysis. Their analysis showed an insignificant impact between electricity consumption and the quality of life.

Table 4
System GMM regression results (energy consumption and child mortality).

Variables	Infant Mortality Rate (MINF)				Under-5 Mortality Rate (MUN5)			
	1	2	3	4	5	6	7	8
MINF(-1)/MUN5(-1)	**0.947*** **	**0.953*** **	**0.957*** **	**0.958*** **	**0.976*** **	**0.948*** **	**0.964*** **	**0.961*** **
	(0.030)	**(0.033)**	**(0.024)**	**(0.028)**	**(0.019)**	**(0.026)**	**(0.025)**	**(0.023)**
Energy consumption	0.006	**−0.051** **	−0.024	**−0.049** **	−0.028	**−0.040* **	−0.014	**−0.044** **
	(0.013)	**(0.024)**	(0.018)	**(0.020)**	(0.019)	**(0.020)**	(0.027)	**(0.018)**
Carbon emission		**0.033*** **		**0.033*** **		**0.028* **		**0.032*** **
		(0.009)		**(0.009)**		**(0.015)**		**(0.009)**
Total natural resource rent			**−0.007** **	**−0.007** **			**−0.012*** **	**−0.010** **
			(0.003)	**(0.003)**			**(0.004)**	**(0.003)**
GDP per capita	**−0.061** **	**−0.053** **	0.020	0.011	0.017	0.008	0.024	0.005
	(0.017)	**(0.022)**	(0.012)	(0.015)	(0.011)	(0.021)	(0.017)	(0.023)
Govt. Expenditure on education	**−0.004** **	**−0.003* **	−0.0025	**−0.005** **	−0.002	**−0.003* **	−0.002	**−0.003** **
	(0.002)	**(0.002)**	(0.002)	**(0.003)**	(0.001)	**(0.002)**	(0.002)	**(0.0014)**
Health expenditure	−0.007	−0.002	−0.002	0.001	−0.003	−0.005	−0.004	−0.004
	(0.004)	(0.005)	(0.003)	(0.005)	(0.004)	(0.004)	(0.004)	(0.005)
Institutional quality	0.012	**−0.042** **	**−0.023** **	**−0.052** **	**−0.019*** **	**−0.027** **	−0.025	**−0.053** **
	(0.013)	**(0.020)**	**(0.010)**	**(0.019)**	**(0.002)**	**(0.012)**	(0.015)	**(0.026)**
Inflation rate	**0.002*** **	**0.002*** **	**0.002*** **	**0.002*** **	**0.001*** **	**0.001*** **	**0.001*** **	**0.001*** **
	(0.0001)	**(0.0001)**	**(0.0001)**	**(0.0001)**	**(0.0004)**	**(0.0004)**	**(0.0004)**	**(0.0004)**
Urban population rate	−0.003	−0.003	−0.009	−0.005	−0.007	−0.009	−0.009	−0.004
	(0.008)	(0.008)	(0.010)	(0.010)	(0.007)	(0.011)	(0.013)	(0.010)
Improved sanitation facilities	**−0.002** **	−0.001	**−0.0016* **	**−0.0017** **	**−0.001** **	**−0.002** **	**−0.002** **	**−0.0018* **
	(0.001)	(0.001)	**(0.001)**	**(0.001)**	**(0.0005)**	**(0.001)**	**(0.001)**	**(0.001)**
Improved water sources	−0.000	0.001	0.000	0.000	0.001	−0.000	0.000	0.000
	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)	(0.001)
Constant	**0.270** **	**0.468** **	**0.208* **	**0.353*** **	**0.145** **	**0.463** **	0.111	**0.379* **
	(0.132)	**(0.236)**	**(0.120)**	**(0.108)**	**(0.073)**	**(0.208)**	(0.097)	**(0.190)**
AR (1)	(0.017)	(0.022)	(0.036)	**(0.067)**	(0.018)	(0.011)	**(0.064)**	(0.029)
AR (2)	**(0.485)**	**(0.324)**	**(0.469)**	**(0.423)**	**(0.432)**	**(0.345)**	**(0.219)**	**(0.404)**
Sargan OIR	**(0.403)**	**(0.445)**	**(0.547)**	**(0.453)**	**(0.452)**	**(0.428)**	**(0.402)**	**(0.420)**
Hansen OIR	**(0.997)**	**(0.998)**	**(1.000)**	**(1.000)**	**(0.999)**	**(1.000)**	**(1.000)**	**(1.000)**
Fishers	**6838.6*** **	**6397.8*** **	**8999.7*** **	**5415.3*** **	**7192.3*** **	**5806.7*** **	**2486.6*** **	**4217.3*** **
Instruments	15	17	17	19	15	17	17	19
Countries	23	23	23	23	23	23	23	23
Observations	197	197	197	197	197	197	197	197

Note: Standard errors in parentheses; ***p < 0.01, **p < 0.05, *p < 0.10; MINF is mortality rate, infant (per 1000 live births); and MUN5 is under 5 mortality rate (per 1000 live births). The bolded values signify significance of (a) estimated parameters and F-statistics and (b) failure to reject the null hypotheses of: (i) no autocorrelation in the AR(1) & AR(2) tests and; (ii) the validity of the instruments in the Sargan OIR test.

Khandker et al. (2013) considered the welfare impacts of rural electrification in Vietnam using a panel survey data for two project years under the first World Bank financed Vietnam Rural Energy Project initiated in 2000. Using a panel estimation technique, the study shows the effect of electrification on household income, expenditure and children's education in Vietnam and concluded that electricity has welfare impact on household income through the farm and non-farm income channel. The study contributed to literature by showing testing for the development outcome and quantifying the benefit outcome. However, the results show evidence of endogeneity in the analysis, which suggests the appropriate panel estimation procedure was not followed.

Bridge et al. (2016) examined the effect of electricity on income and quality of life in Nepal, using the Nepal Living Standard Survey III conducted between 2010 and 2011. Employing a simultaneous equation via the three State Least Square method, with the hope of accounting for endogeneity issues, the study reported a positive but not significant effect of electricity consumption on health. In addition, the study shows that there is a highly significant magnitude of electricity impact on the quality of life via household income, education attainment and agricultural productivity. The study suffers from an inappropriate measure of the variable considered for the analysis, suggesting that better results can be achieved with further research progress in the country. Studies in China have shown that electricity access has been successfully integrated into rural development and that biogas usage for cooking has been reported compatible with sustainable development (Bhattacharyya, 2012).

Contrariwise, some studies have argued that the present quantity and quality of electricity energy consumed has a negative impact on the quality of life. For example, Jorgenson et al. (2014) developed a framework for estimating effects of energy consumption on wellbeing fundamentally by reflecting economic development in Central and Eastern European Nation between 1992 and 2010. The study analyses the measure of wellbeing indicators and energy consumption as well as measures of energy intensity. They further quantified the adjusted ratio of energy consumption and the average life expectancy rate and reported that energy consumption negatively impacts on life expectancy during the period. The results by these authors indicated that economic development is not sustainable due to the perceived negative estimated coefficient on social wellbeing indicators.

Relatively few empirical works on the relationship between energy use and life expectancy find mixed results. For instance, Wang and Luo (2018) considered the relation between life expectancy and energy utilization in China using the household quality database and reported that electricity utilization exhibits different relations with life expectancy in China. The study reported a positive correlation between household energy consumption and life expectancy at birth, while the use of domestic coal and biomass fuel shows a significant negative relation with life expectancy, central heating, residential boilers; moreover, it shows that natural gas lacks any correlation with life expectancy. It is worth noting that the evidence documented in this study is contrary to the earlier report by (Bhattacharyya, 2012) in China.

Yang and Chen (2014) conducted an ecological economic evaluation of the Manwan Hydropower Plant in the middle reach of the Lancang River. They quantified the variation of river ecosystem service functions as it affects quality of life and reported that hydropower construction has

impaired the function of river ecosystem services based on the quantitative analysis of both positive effects and negative effects. An indicator system was also established to evaluate the effects of hydropower construction in terms of ecosystem service valuation methods.

Mattmann et al. (2016) estimated a meta-analysis of existing research related to the economic valuation of the external effects of hydropower. A database consisting of 81 observations derived from 29 studies valuing the non-market impacts of hydropower electricity generation is constructed with the main aim of quantifying and explaining the economic values for positive and negative hydropower externalities. The authors reported significant evidence for public aversion towards deteriorations of landscape, vegetation and wildlife caused by hydropower projects.

Zhang et al. (2016) compared the effects of long-term suburban growth on travel behavior, energy consumption, and GHG emissions through a case study of neighborhoods in central Phoenix and the suburban city of Gilbert, in the Phoenix metropolitan region of the United States. Motorized travel patterns in these study areas are estimated using 2001 and 2009 National Household Travel Survey data. Energy consumption and GHG emissions, including both Carbon Dioxide and Nitrous Oxide for each study area, are estimated based on the corresponding trip distribution results.

Equally, Sweidan and Alwaked (2016) empirically investigated the effect of economic development on the energy intensity of human well-being (EIWB) in the Gulf Cooperation Council (GCC) during the period 1995–2012. Employing a time-series cross-sectional Prais–Winsten regression model with panel-corrected standard errors (PCSE), they reported that economic development has had a significant positive impact on EIWB during the period. Their results indicated that the stress on the environment had increased in a noticeable manner from 1995 until 2006.

Apparently, the various studies discussed above include scholars work in different environments, use different data, and apply different methods reflects their various diversity of opinions and submissions of the studies presented thus far. This makes it difficult to conclusively conclude that the appropriate answers have been found in relations to the energy-life hypothesis. One night be tempted to argue that the intellectual debate is capable of generalization, but the contrasting evidence documented, and conclusions reached from the studies reviewed above have triggered more problems that demand urgent inquiries. Therefore, this study will not examine the same variables as these studies; rather, its empirical results will be quite different.

3. Methodology

The theoretical foundation of this study is based on Gary Becker's human capital hypothesis, while the model adapted follow the Grossman Demand for healthcare modelling strategy. The considerations and choice of the theory and model is based on their ability to explains the basis of conservative energy policy as it elucidates the rationale to preserve and extend human life for future economic progress. In literature the Grossman demand for healthcare is applied when consumption related goods are likely to have negative externalities as in the case of energy-health nexus. Thus, the baseline model for this study is specified as follows:

$$mrt_{i,t} = \varphi_0 + \varphi_1 ec_{i,t} + \varphi_2 tnat_{i,t} + \varphi_3 ep_{i,t} + \Phi CVR_{i,t} + v_{i,t} \quad (1)$$

where: *mrt* represents a vector of mortality rate (infant (*minf*) and under-five (*mun5*)); *ec* denotes energy use per capita; and *tnat* is total natural resources rents. *CVR* is a vector of control variables, which are government spending on education (*gedu*), public health expenditure (*hexp*), gross domestic product per capita (*gdppc*), inflation rate (*inf*), improved sanitation facilities (*isf*), improved water source (*iws*), urbanization rate (*urban*) and aggregate institutional index (*insq*). The institutional quality consists of six indices, namely control of corruption, voice and accountability, rule of law, government effectiveness, regulatory quality and political stability. It ranges from −2.5 (being the weakest) to 2.5 (being the strongest). Meanwhile, φ_0, φ_{1-3} are coefficients, Φ– a vector of parameters of our control variables, i represents country, t is time and v is the stochastic term with expected zero mean and constant variance. Similarly, a pooled OLS and a system of generalized method of moments were employed for empirical estimation.

The paper uses annual panel data that comprises of 23 African countries[1] from 1999 to 2014. The choice of data and period examined were based on the availability of data. This selection was influenced by the data available for energy use per capita. Like previous studies, the variables used were sourced from World Bank database – World Development Indicators. The study verifies four facts in existing Africa socio-economic data. First, it is expected that the effect of energy consumption is twofold on child mortality: either positive or negative. Specifically, it will have positive effects if the negative externalities, in terms of pollution, on child mortality and the technology for exploring energy is not environmentally friendly. Besides, it will reduce child mortality as proceeds gotten from increasing energy use will be used to enhance human welfare, including low infant or under-5 mortality rates. Second, the total natural resources rent will have an indirect relationship with the child mortality rate. This means that income received from natural resources exploration will be used towards the betterment of people including children living in the country. However, a direct relationship is expected between carbon emission and child mortality. It implies that environmental hazards constitute a great challenge to human healthiness including infants and under-5 children.

Third, it is expected that a high-income level will reduce the child mortality rate. It thus implies a negative relationship between GDP per capita and child mortality. Moreover, an indirect relationship is anticipated from public education and health spending to the infant and under-5 mortality rate. This means that in a society where the people have quality education and healthcare facilities, mortality rate of infants and under-5 years will reduce. Fourth, it is expected that a country with high female secondary enrolment will have low rate of child mortality. As for inflation rate, we expect a high rate of child mortality due to the increasing cost of living which makes healthcare services costly. In addition, the study hypothesized an indirect relationship between inflation and child mortality. For the rest of the analysis, the study presumes that a negative relationship exists between improved sanitation facilities and improved water source, on the one hand, and child mortality, on the other. It means that a society with better sanitation and water sources will experience enhanced healthy living as well as low child mortality. This study also assumes a direct relationship between urbanization and child mortality. Thus, an increasing rate of urbanization stresses the available healthcare facilities and resources in urban centers as they lack ability to control child mortality. Finally, an indirect link is presumed between institutions and child mortality. It means that a stable institutional framework guarantees human welfare development by ensuing that people receive quality healthcare services capable of reducing infant and under-5 mortality rates.

4. Empirical results

In this section, we present the results of the preliminary analysis of the variables consider before the empirical findings using pooled OLS and system GMM approaches is discussed. Tables 1 and 2 show the results of the descriptive statistics and correlation analysis of the variables. The mean values of infant and under-5 mortality rate (per 1000 live births) stood at 63.52 and 99.84, respectively. It implies that for every 1000 live births recorded in the sample set of our selected countries, there is 63.52

[1] They are Angola, Benin, Botswana, Cameroon, Congo Dem. Rep., Congo Rep., Cote D'Ivoire, Eritrea, Ethiopia, Gabon, Ghana, Kenya, Mauritius, Mozambique, Namibia, Niger, Nigeria, Senegal, South Africa, Tanzania, Togo, Zambia, Zimbabwe.

and 99.84 deaths of infants and under-5 years children. The average values of energy use, public health and education expenditure, GDP per capita, CO_2 emission, natural resource rent, institutions, inflation and urbanization rate remain as reported in the previous sub-section. Lastly, the mean percentage of people who have access to improved sanitation and water sources are 31.06% and 67.33%, respectively.

Table 2 shows the correlation matrix and reveals that the two infant mortality indices have negative correlation with energy consumption per capita. Likewise, human capital investment indicators (education and health) have negative coefficients, suggesting that an increase in public health and education spending will lead to a decrease in child mortality rate in SSA countries. A glance at the scatter graphs of the links between the variables further confirmed the results of the correlation analysis. Moreover, the results of other covariates were also reported in the table. Meanwhile, the correlation coefficients are low, implying the suitability of the variables for the analysis and the absence of any multi-collinearity problem.

Fig. 3 shows the nature and pattern of air pollution in the selected Africa countries examined. A glance at the figures presented shows the nature and patterns of air pollution among the countries. Clearly, it becomes glaring to note that lack of clean energy and the resultant effect of air pollution are major factors that impact increased mortality rates in Africa. Furthermore, evidence shows that the major driver of environmental risk to health in Africa is the resultant effect of energy related crises. As observed, air pollution has made Africa a major candidate of acute health related cases as compared to other continents around the world. While the power and energy crises persist in Africa with actions being taken by respective African countries supported by international bodies and agencies to resolve the crises, the reported gain in achieving success seem slow. Part of the explanation is that electricity shortage, poor waste management, and heavy reliance on biomass fuel and coal have far more negative effects on human life.

Fig. 4 shows the connection between air pollution and child mortality rates in Africa. Interestingly, the figure indicates that increases in child mortality rates are due to exposures to waste energy related factors. However, it is difficult to assume that energy resources account for the overall exposure to infant mortality rates and all pollution related crises. At best, available figures confirmed that more than 53% of infant mortality rates could be traced to poor environmental quality. There is also the pointer that increased cases of health-related risks such as pneumonia, non-communicable diseases as well as lung cancer could be attributed to the absence of quality energy use in the region. This therefore calls for urgent intervention that should be approached holistically to reduce the incidences of power/energy environmentally related diseases. It is obvious from the figures that the much-aspired socio-economic development for the African region may remain a mirage until the problem of power and clean energy is addressed and made available and accessible to all.

This section presents the main results of the regression analysis, which is based on the pooled OLS and *system*-GMM estimators. Also, the discussion and finding are discussed extensively.

Table 3 shows the baseline results of a pooled OLS of the child mortality regression. The results broadly indicate that the adjusted R-square of about 70% of the total variability of mortality rates is explained by the model. Column 2 of Table 3 shows the estimated results of the correlates of infant mortality rates and energy related predictors. The results show that a 1% point increase in energy consumption implies a decrease in infant mortality rates by more than 10%, while a 1% point increase in carbon emission implies an increase in infant mortality rate of more than 6%, holding other variables constant. Similarly, Column 6 of Table 3 shows the baseline estimated pooled regression results on the effects of energy related predictors on under-5 mortality rates. The results indicated that a 1% point increase in energy consumption implies a decrease in under-5 mortality rates of 26%, while 1% point increase in carbon emission implies an increase in under 5 mortality rates of more than 9%, holding other variables constant.

The outcome of the analysis suggests that: (a) on average, the child mortality rate of both infant and under-5 years are negatively and significantly related with energy consumption in SSA countries; (b) carbon emission increases the adverse effect of child mortality, whereas natural resources rents reduce the undesirable effect of the infant and under-5 mortality rate; (c) human capital investment, most especially through public education expenditure, has the potentials of reducing child mortality; (d) other factors capable of reducing infant mortality rate are institutions, improved sanitation facilities and water sources. However, the coefficient of income per capita was positive, implying that the low level of income earned by majority of the citizens in these countries is sufficient to cater for the needs of their families, most especially reducing the child mortality rate.

In addition, Table 4 presents the estimates of the *system* GMM regression results. The dependent variables are infants and under-5 child mortality rate. The results broadly indicate that both current and past values of the dependent variables have a positive important implication on its current variation within the African countries examined.

Findings further maintained that, on average, energy consumption had negatively influenced child mortality for infants and those under the age of 5 years. Column 2 of Table 4 shows that a 1%-point increase in fraction of energy consumption implies a decrease in infant mortality rate of 5%, while a 1% point increase in the fraction carbon emission implies an increase of 3.3% in infant mortality rate, holding other variables constant. Comparably, a 1%-point increase in the fraction energy consumption implies a decrease in under-5 mortality of 4%, while a 1% point increase in the fraction of carbon emission implies an increase in under-5 mortality rates of 4%. Similarly, the coefficients of carbon emission and total natural resources rent is similar to the findings derived under the pooled OLS approach. A high pollution rate is found to cause rising mortality rates, whereas proceeds from natural resource rents reduce the level of child mortality. Likewise, the coefficients of public education spending follow the results of our pooled OLS estimates. Other indicators that significantly reduce child mortality rate are institution and improved sanitation facilities.

Meanwhile, the lag one of mortality rates (infant and under-5), which were positive and lower than one and statistically significant at 5%, indicate that convergence condition in the selected countries in Africa. Also, the Fishers tests were found to be significant at a 0.05 critical value, implying that all the coefficients are jointly significant at the conventional level. For the post estimation test, we did not reject the null hypothesis of the first order of serial correlation [AR (1)], while the null hypothesis of the second order of serial correlation was not accepted. This is consistent with the findings of Arellano and Bond (1991); Hansen (2012); Hayakawa (2014); Kiviet (2016); Moral-Benito et al. (2019) to be rejected for the instruments to be valid. This suggests that the instruments are valid and do not correlate with the disturbance term.

5. Conclusion

The article examined the effect of energy use on the infant mortality rate in Africa over the period 1999 and 2014. The Gary Becker human capital hypothesis and Grossman demand for health model were used to interact energy predictors with mortality rate fundamentals. Empirical evidence is based on the pooled OLS and system GMM estimators. The results suggest that infant mortality rate of both infant and under-5 years are negatively and significantly related with energy consumption in the region. Also, an increase in carbon emission has a consequential effect on the infant mortality rate. Further results indicated that human capital investment, most especially through public education expenditure, has the potential to reduce infant mortality rates. Thus, the study recommends that the provision of clean, reliable and affordable energy is needed to improve infant mortality rates in the region. In addition, an improvement in health expenditure and energy education is required to enhance mitigation of carbon and improve energy savings. Furthermore, the study makes an important contributed to the existing literature by

addressing the fundamental causes of infant mortality rate in Africa in many ways. First, the study interacts the energy industry and health sector to explain the causes of infant mortality rates in Africa. Second, it provides empirical evidence based on the system of generalized method of moments (system GMM). Third, the result supports the Gary Becker hypothesis that suggested that economic factor input plays a predominant role in the ability of an individual to preserve life. More importantly, the results of the system-GMM are able to eliminate potential bias due to reverse causality in the estimation. This study is supported by the findings in Alola et al. (2019) and Shobande (2019) but questioned the findings by Faisal et al. (2018). Finally, making empirical progress requires that future studies focus on the effect of disaggregated renewable energy consumption on life expectancy in Africa. Such cross-country level studies are essential for more targeted policy suggestions on health-related crises.

Declaration of competing interest

The author declares no potential conflicts of interest with respect to the research, authorship, and/or publication of the article.

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